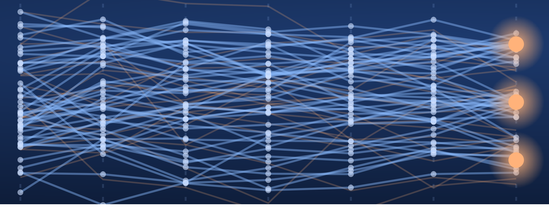
NeurIPS 2026 REINFORCEMENT LEARNING · GENERATIVE POLICIES

Generative Actor-Critic with Soft Bridge Policies

One step outside, a **soft bridge** inside: a principled MaxEnt actor that beats many-step diffusion in a single pass.

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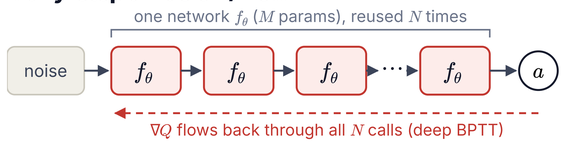
Code (JAX)
Scan for the repo
github.com/skypitcher/soft_gac_jax
arXiv:2605.08733

1 Two problems

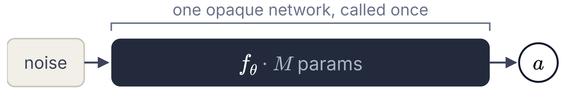
Why generative? Diffusion and flow policies capture **multimodal, non-Gaussian** actions (e.g. go left *or* right around an obstacle; a Gaussian's single peak sits in the middle). But they face two problems.

① **Cost vs. performance.** Many-step samplers are strong but call one network N times. A **one-step** generator is fast, but returns usually drop.

Many-step diffusion / flow



One-step generator (black box)

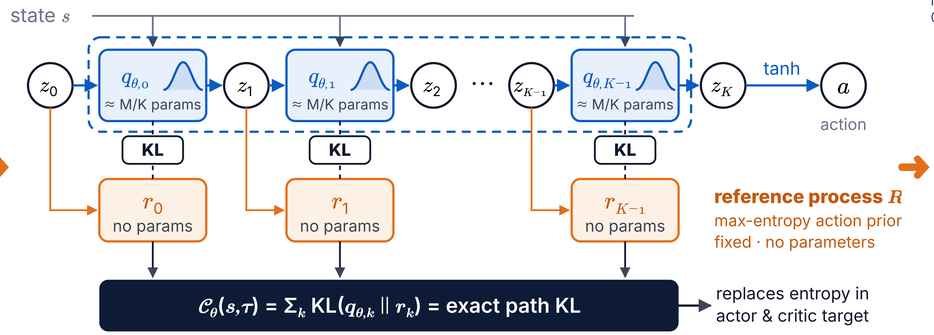


② **Intractable entropy.** For **any implicit generator** (one step or many), $\pi_\theta(a \mid s)$ cannot be computed. Prior soft regularizers **approximate** the entropy (inexact) or use **surrogates** (indirect): neither guarantees endpoint-equivalence.

Can one policy be truly soft, fast, and better?

2 Open the box: a soft bridge

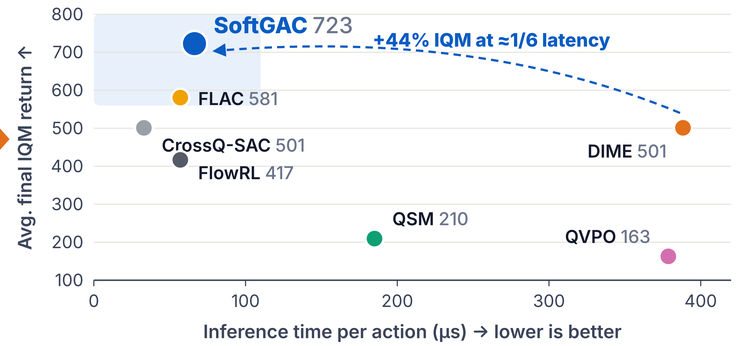
About the **same M params**, still **one call** (the dashed box is f_θ). Inside, the budget is split into K small Gaussian layers (**≈ M/K params** each), each matched by one step of a fixed max-entropy reference process (no params). Their KLs are closed form and add up to the **exact** soft regularizer, computed in the same pass.



3 Faster, yet stronger

+44%
avg. IQM over the strongest many-step actor (DIME), with 0.43M vs. 0.44M actor params

≈1/6
of DIME's per-action inference latency



Mean final IQM on the 8 hard tasks vs. per-action latency (mean of 3 domains, M3 Pro CPU), from Figs. 3 and 9 of the paper. Also +24% over the best baseline overall (FLAC).

4 Why it is principled: MaxEnt in path space

SAC regularizes the action law: $\mathbb{E}_\pi[Q] - \alpha D_{\mathrm{KL}}(\pi \,\|\, u)$. For any implicit generator, $\pi_\theta(a|s)$ is intractable, so prior methods approximate the entropy (inexact) or use surrogates (indirect). We regularize the **whole path** $\tau = (z_0, \ldots, z_K)$ instead:

$$\mathcal{J}_{\mathrm{path}} = \mathbb{E}_{\tau\sim P_\theta}\big[Q(s, \tanh z_K)\big] - \alpha\, D_{\mathrm{KL}}\big(P_\theta(\cdot \mid s) \,\|\, R\big)$$

PROP. 1 · PATH KL CONTAINS SAC'S REGULARIZER

$$D_{\mathrm{KL}}(P_\theta \| R) = \underbrace{D_{\mathrm{KL}}(\pi_\theta \| u_R)}_{\text{endpoint KL (SAC)}} + \underbrace{\mathbb{E}_{a\sim\pi_\theta}\big[D_{\mathrm{KL}}(P_\theta(\cdot|a) \| R(\cdot|a))\big]}_{\text{how } a \text{ is generated}}$$

IDEAL THM. 1 · UNCONSTRAINED BASE
Over all path laws $P \ll R$, *including* the law of z_0:

$$\frac{dP_Q^\star}{dR}(\tau \mid s) \propto e^{Q(s,\tanh z_K)/\alpha} \implies \pi_Q^\star(a \mid s) \propto u_R(a)\, e^{Q(s,a)/\alpha}$$

ENDPOINT-EQUIVALENCE With uniform u_R, $\pi_Q^\star$ is **exactly SAC's Boltzmann policy**. The path term only shapes *how* each action is reached.

PRACTICAL THM. 2 & COR. 1 · FIXED BASE
The ideal optimum also reweights z_0 by $Z_Q(s, z_0)$, a state-dependent base sampler. A real actor fixes p_0 (ours: q_{ref}), so only the transitions are tilted:

$$P^\star_{Q,p_0}(d\tau \mid s) = p_0(dz_0)\, \frac{e^{Q(s,\tanh z_K)/\alpha}}{Z_Q(s,z_0)}\, R(d\tau_{1:K} \mid z_0)$$

The base adds only a constant, and its gap $\Delta(s) \geq 0$ bounds the endpoint error: $D_{\mathrm{KL}}(\pi^\star_{Q,p_0} \| \pi^\star_Q) \leq \Delta(s)/\alpha$.

5 How it is built and trained

Each layer, and its reference step

$$q_{\theta,k}(z_{k+1} \mid z_k, s) = \mathcal{N}\big(z_k + h\, u_{\theta,k}(s, z_k),\ 2h \operatorname{diag} \sigma^2_{\theta,k}\big)$$

$$z_0 \sim q_{\mathrm{ref}} \propto \textstyle\prod_i \operatorname{sech}^2 z_i, \qquad r_k = \mathcal{N}\big(z_k - 2h \tanh z_k,\ 2hI\big)$$

$h = 1/K$. Each layer is its own small MLP (≈ M/K params; about M in total), not a shared sampler. $\tanh$ maps q_{ref} to the uniform law, so R is a max-entropy action prior.

LEMMA 1 · EXACT CONTROL ENERGY

$$\mathcal{C}_\theta(s,\tau) = \sum_{k=0}^{K-1} \frac{1}{2} \sum_{i=1}^{d_a} \left[\sigma^2_{\theta,k,i} + \frac{(\mu_{\theta,k,i} - \mu_{R,k,i})^2}{2h} - 1 - \log \sigma^2_{\theta,k,i}\right]$$

with $\mathbb{E}_{\tau\sim P_\theta}[\mathcal{C}_\theta] = D_{\mathrm{KL}}(P_\theta \| R)$: closed form, no estimation.

Drop-in soft actor-critic

Endpoint entropy is replaced by control energy, used **directly** in both updates:

$$y = r + \gamma(1-d)\Big[Q^{\min}_{\bar\phi}\big(s', \tanh z'_K\big) - \alpha\, \mathcal{C}_{\bar\theta}(s', \tau')\Big]$$

$$\mathcal{L}_{\mathrm{actor}}(\theta) = \mathbb{E}_{s,\ \tau\sim P_\theta}\Big[\alpha\, \mathcal{C}_\theta(s,\tau) - Q^{\min}_\phi\big(s, \tanh z_K\big)\Big]$$

PROP. 2 · A REVERSE-KL PROJECTION

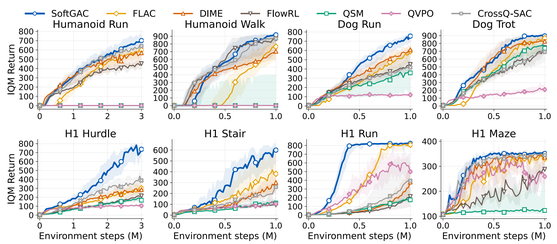
$$\mathcal{L}_{\mathrm{actor}}(\theta \mid s) = \alpha\, D_{\mathrm{KL}}\big(P_\theta \,\|\, P^\star_Q\big) + \mathrm{const}$$

α tracks a control budget $\mathcal{C}_{\mathrm{target}} = \rho_{\mathrm{ctrl}} K d_a$. Deployment follows the mean bridge path.

6 Results: faster does not have to mean weaker

DMC (Humanoid, Dog) + HumanoidBench (H1) · 12 tasks · 8 seeds · IQM, 95% CI · all actors re-implemented in one codebase with comparable parameter budgets.

Learning curves on the 8 hard tasks

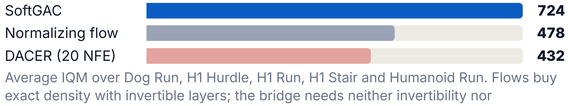


Largest gains on high-dimensional, long-horizon locomotion (Humanoid Run, Dog Run, H1).

vs. 20-step diffusion actors · H1 Hurdle (3M)

Actor	NFE	Params	IQM	Inference	Train
SoftGAC	**1**	**0.34M**	**737**	**63 μs**	**6.0 h**
DACER	20	0.32M	377	400 μs	7.6 h
SDAC	20	0.32M	60	410 μs	9.3 h
MaxEntDP	20	0.40M	18	509 μs	24.9 h

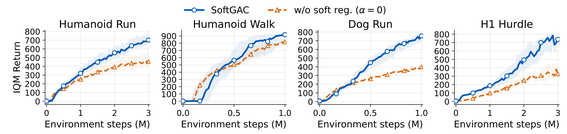
vs. a one-step flow with exact density



Average IQM over Dog Run, H1 Hurdle, H1 Run, H1 Stair and Humanoid Run. Flows buy exact density with invertible layers; the bridge needs neither invertibility nor Jacobians.

7 Why it works

The soft path term matters



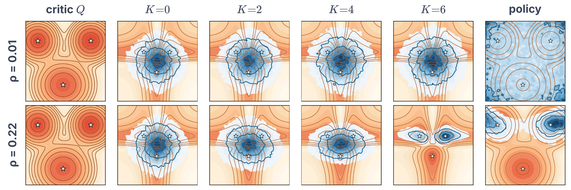
Same actor and critic with $\alpha = 0$: removing the control energy consistently hurts.

Analytical KL » estimated entropy

Path KL (ours)	729
GMM entropy	349

Same actor with DACER's GMM endpoint-entropy estimator. Avg. IQM, 4 tasks.

The budget trades coverage for value



2D toy, 3 value modes. Real policies are multimodal: **53.8%** of H1 Hurdle states (CrossQ-SAC: 0%). Also robust to base, depth and critic (XQC).

Takeaways

① **Soft**
Analytical path-KL, endpoint-equivalent to SAC. No estimates or surrogates.

② **Fast**
One call per action, about the same params: ≈1/6 the latency.

③ **Better**
+44% return over the strongest many-step policy.

Limitations: extra design choices; state-based simulation only.

40th Conference on Neural Information Processing Systems (NeurIPS 2026)

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